intro of big data

Fundamentals of Data Engineering

Trainer: Pradnyaa S. Dindorkar

CCAT =>section B => 5 MCQ

4 pm - 7 pm

day 1 26 jun

day 2 => 29 jun

day 3 => Quiz 1 30 jun => 10

=> Quiz 2 1 july => 10

=> end 2 july => 20

poll ,MCQ => 30

2 mock 10

80+

BDA

Pre-requistes

1: appln dev / programming

2: database

3: Networking

4: OS (Linux)

Introduction syllabus

- **Big Data Fundamentals**
  - Evolution of Data Enggineering | V's: Volume, Velocity, Variety, Veracity, Value
- **Databases**
  - RDBMS - ACID, SQL (basic concept only) | NoSQL - BASE, CAP theorem

day 1

- **Data warehouse - OLAP vs OLTP**
  - Data cleansing, Data transformations and Data modelling | Data warehouse vs Data mart
- **Data Engineering Life Cycle**
  - Source → Ingestion → Storage → Transformation → Serving
  - Ingestion: ETL vs ELT
  - Storage: Distributed storage, Storage services | Processing: Batch vs Stream

day 2

- **Cloud computing fundamentals**
  - Virtualization, Scaling, Elasticity, Cloud service models, Vendors
- **Big Data Technologies**
  - Frameworks: Hadoop, Hive, Spark, Kafka
  - Applications and Job profiles.

day 3

Data Engineering at a Glance

be4 1970 file i/o

Database & Warehouse

CRUD

1970 =>RDBMS
SQL

@1990 -> DWH

@1980

Internet & DotCom

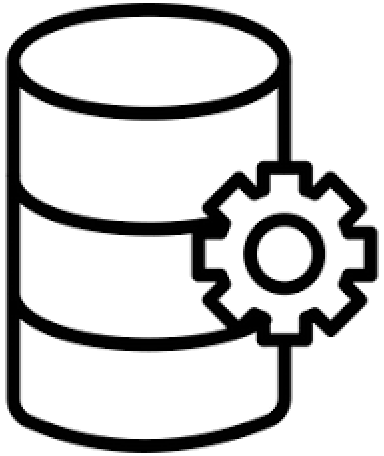
@1995 applet

@2000
data burst

NoSQL Database

@1998

CRUD



massive parallel processing

MPP & Big Data Tech

@2003

1= 7
50=7X50

Cloud Computing

rent

@2010

Big Data characteristics

ola => lat,lng =>2 float x 10sec 3 x 2 X few lakh of cabs

Velocity

Speed at which data is emanating and changes are occurring between the diverse data sets

ola => lat,lng =>2 float

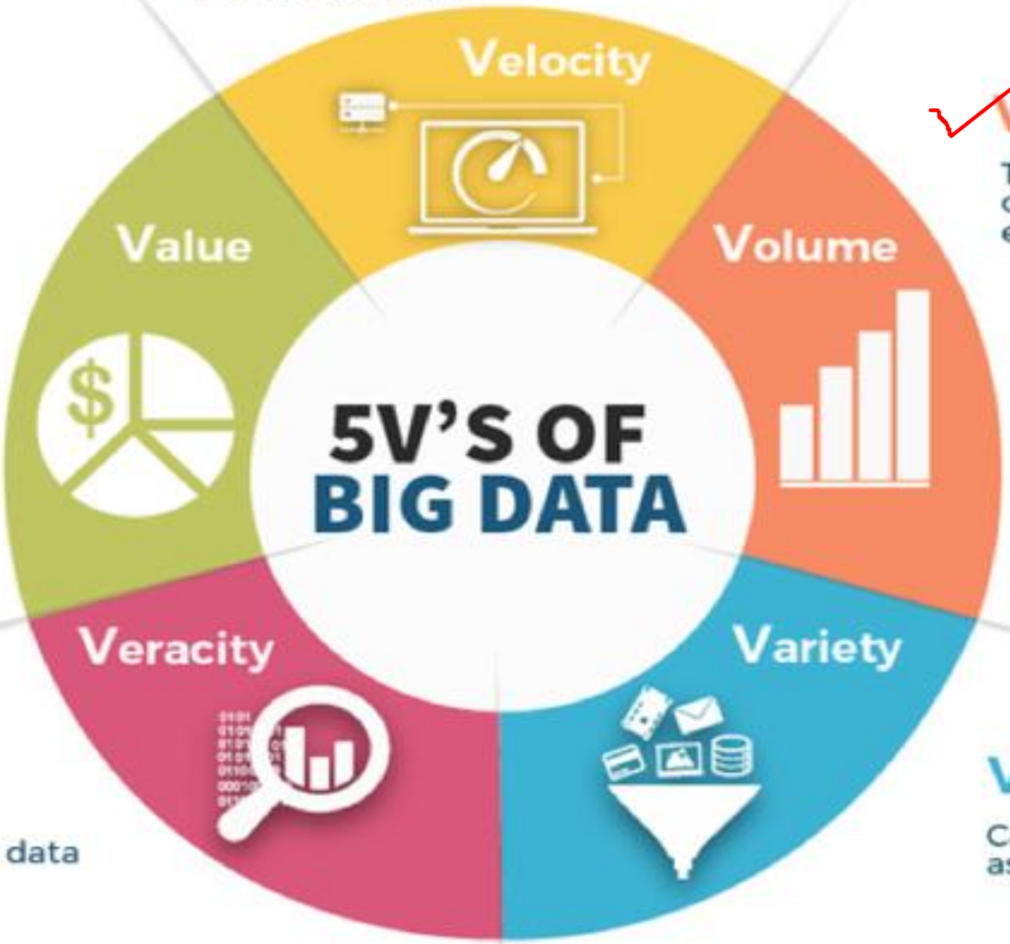
500TB

Volume 100s TB + above

This refers to the sheer volume of data being generated every second.

Value

Having access to big data is all well and good but that's only useful if we can turn it into a value.



Veracity

Data reliability and trust.
Verifying and validating the data

Variety

Can use structured as well as unstructured data.

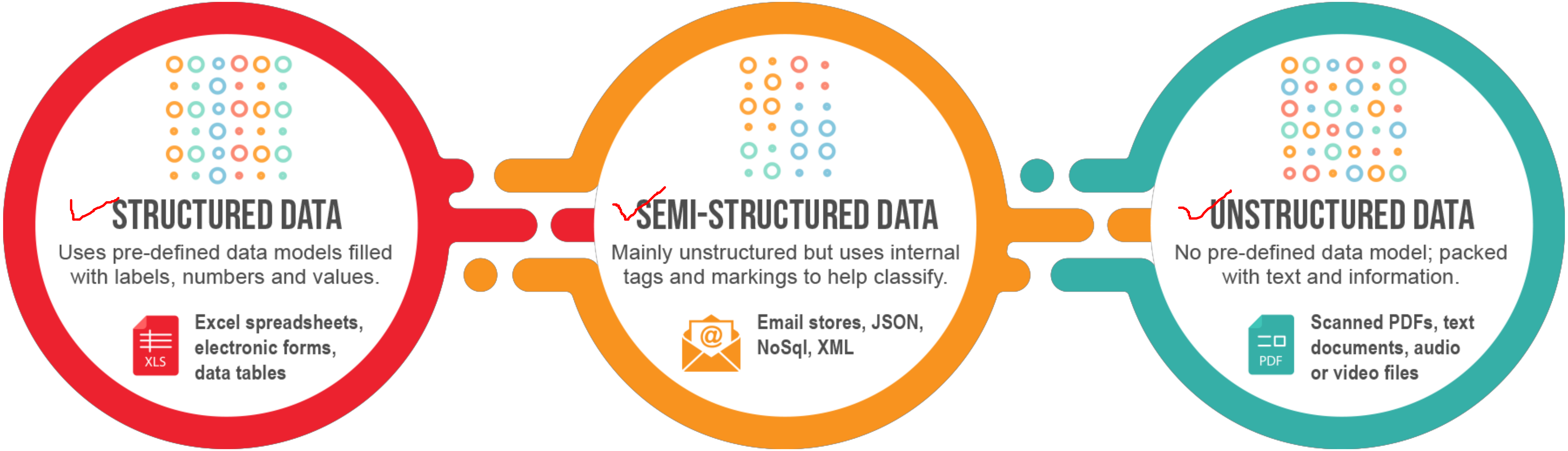
Types of Data

fb => 2008

post : --
links:---
image:---

videos:---
loc:--
tag:---

ML model



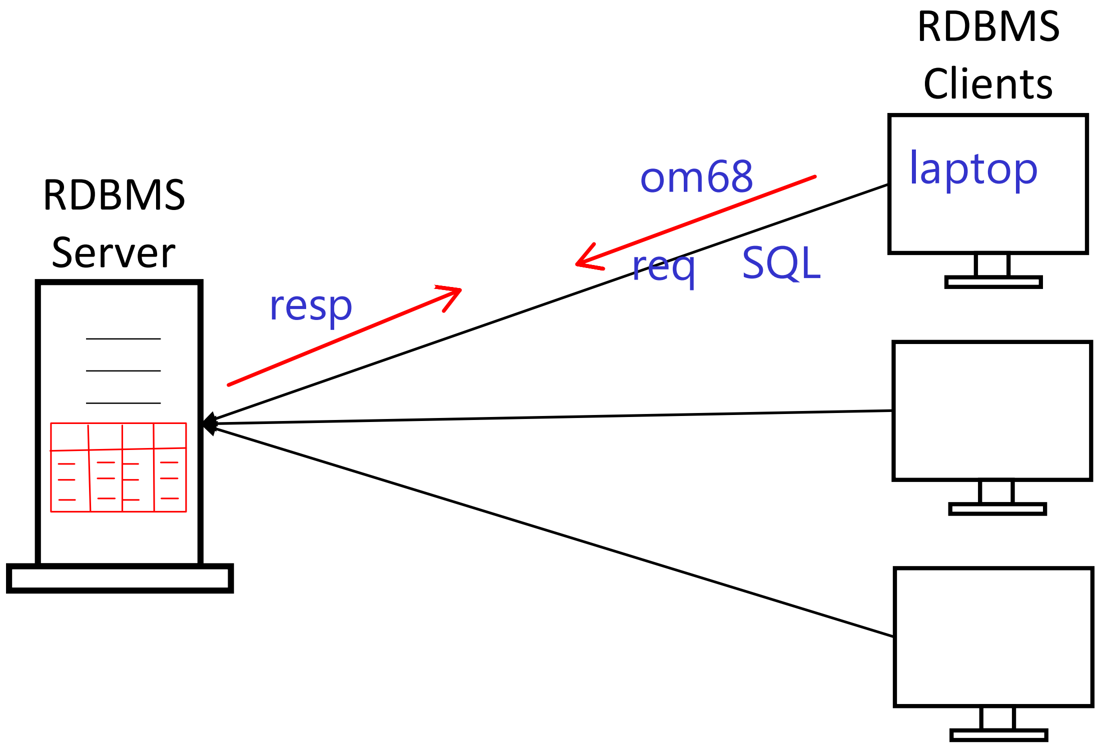
fixed format/
schema

flexible schema

.mp3/4
.png / .jpg

RDBMS

- Every enterprise application need to manage data.
- RDBMS is relational DBMS than manages structured data.
- Data is organized into tables, rows and columns. Tables are related to each other.
- All enterprise RDBMS follow server-client architecture, have built-in relational capabilities, fully ACID transactions, based on Codd's rules.
- DB2, Oracle, MS-SQL, MySQL, Postgre-SQL, MS-Access, SQLite, etc.



SQL – Structured Query language

- **RDBMS data is processed with SQL queries.**
- **ANSI standardised in 1986 and ISO Standardization in 1987.**
- **Five major categories:**
  - **DDL: Data Definition Language e.g. CREATE, ALTER, DROP, RENAME.**
    - CREATE TABLE student(roll INT, name CHAR(40), batchid INT);
  - **DML: Data Manipulation Language e.g. INSERT, UPDATE, DELETE.**
    - INSERT INTO student VALUES(1, 'Ravi', 3);
    - UPDATE student SET name='Ravee' WHERE roll=1;
    - DELETE FROM student WHERE roll=1;
  - **DQL: Data Query Language e.g. SELECT.**
    - SELECT * FROM student; read
  - **DCL: Data Control Language e.g. CREATE USER, GRANT, REVOKE.**
  - **TCL: Transaction Control Language e.g. SAVEPOINT, COMMIT, ROLLBACK.**

Transaction characteristics- ACID

@1970 ==>2026

account

	id	type	balance
papa	1	saving	30000
You	2	saving	5000
	3	current	80000
	4	saving	40000

4000

account

	id	type	balance
papa	1	saving	26000 ~~30000~~
You	2	saving	9000 ~~5000~~
	3	current	80000
	4	saving	40000

4000

update 26000 acc-1

update 9000 acc-2

single

Transaction =>set of DML commands executed as a single unit

either all commands in Transaction are successful or all commands in Transaction are discarded

Atomic =>single unit

Consistent => same result shown to the client

Isolated => several Transaction are done at time without affecting other

Durable =>all changes are saved permanantly

Scalibility

client
work
load
data

5000

2CPU
4GB
1TB

500000

8 CPU
64 GB
32 TB

1: limit

2: SPOF

Single Point Of
Failure

vertical scaling
[up- scaling]

node

4 CPU
16 GB
2 TB

4 CPU
16 GB
2 TB

4 CPU
16 GB
2 TB

4 CPU
16 GB
2 TB

4 CPU
16 GB
2 TB

24 X 7

horizantal scaling
[out - scaling]

500

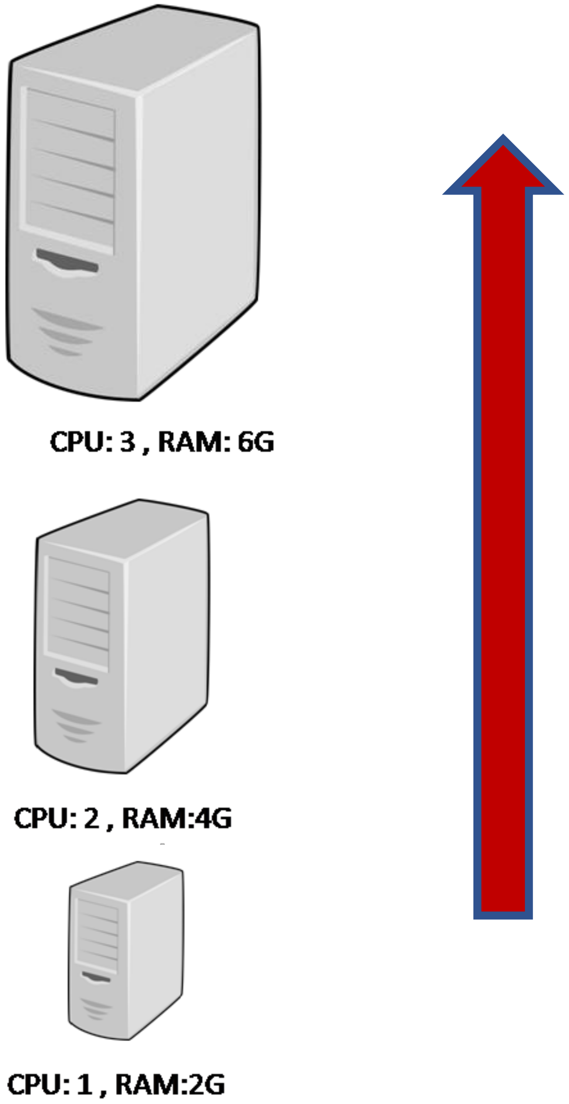
distributed computing

cluster =>set of computer/node connected in a network for dedicated task

Scaling

- Scalability is the ability of a system to expand to meet your business needs.
- Scalability describes a elasticity of the system, ability to adapt to change and demand.
- Good scalability ensures the quality of your service.

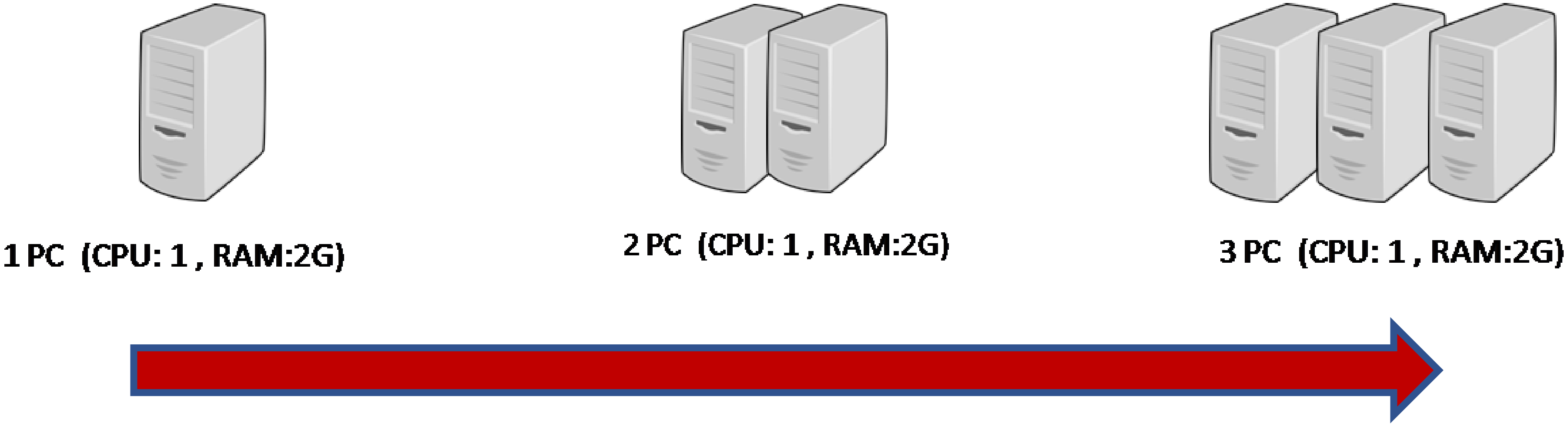
Vertical Scaling



Vertical scaling describes adding more resources to your current machines.

- ✓ increase memory in the system
- ✓ expanding storage by adding hard drives
- ✓ upgrading the CPUs.
- ✓ upgrading network speed.

Horizontal Scaling



Horizontal Scaling refers to adding additional nodes or machines to your infrastructure to cope with new demands.

- ✓ adding a new computer to a distributed software application

NoSQL Databases

@1998 -> Carlo strozzi => NoSQL

- Stands for Not Only SQL @2006-2007
- Manages structured and semi-structured data.
- Prioritizes high performance, high availability and scalability
  quick 24X7
- Designed for Horizontal scaling. Reliable, fault tolerant, Better performance/Speed.
- No declarative query language
- Uses: Huge data (TBs), Many Read/Write ops, Scalable, Flexible schema.
- Don’t use if: Need high consistency, Multiple relations
- BASE transactions and Based on CAP Theorem

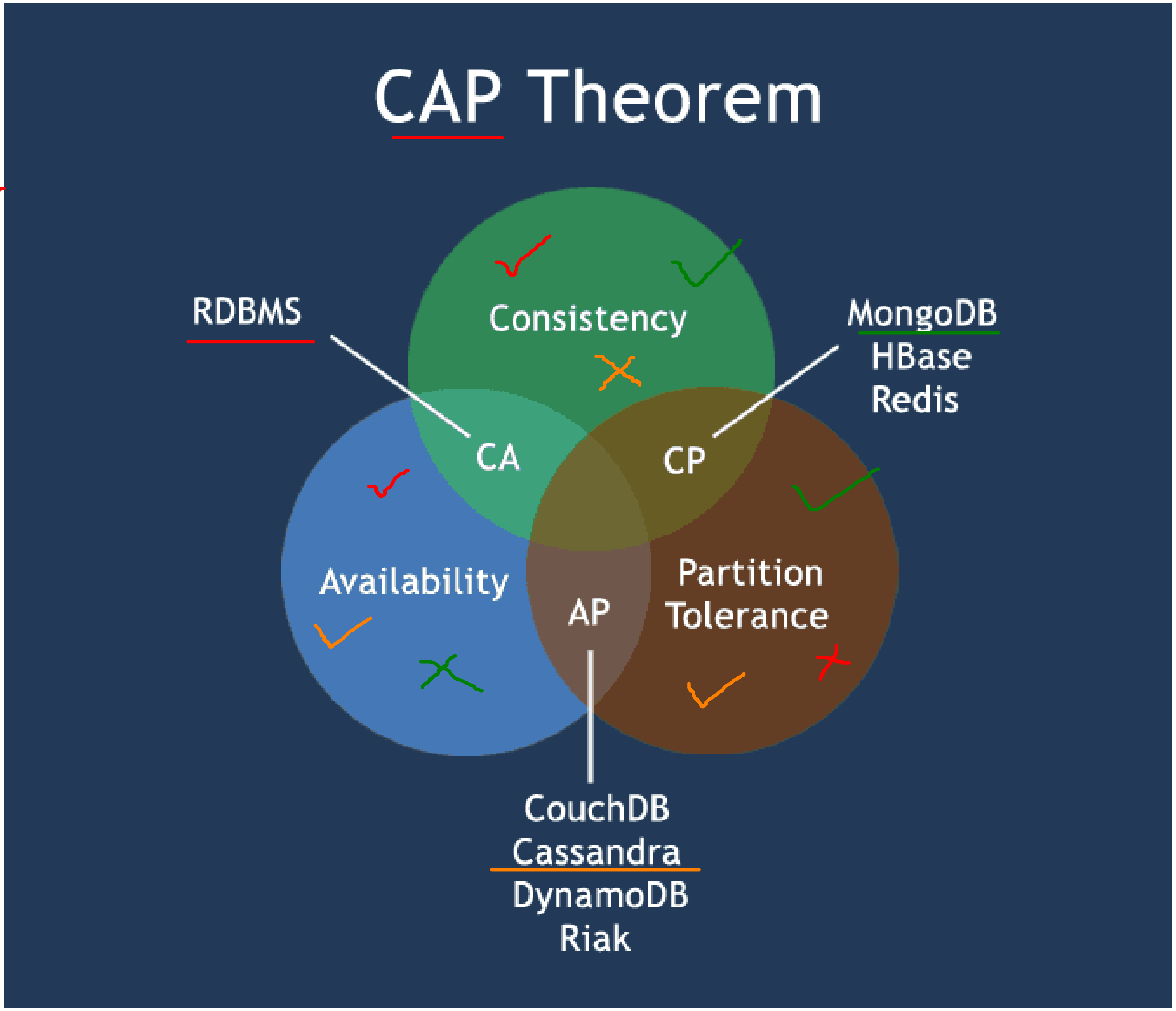
CAP Theorem

Brewer's theorem

- **Consistency - Data is consistent after operation. After an update operation, all clients see the same data.**
- **Availability - System is always on (i.e. service guarantee), no downtime.**

24 X 7

- **Partition Tolerance - System continues to function even the communication among the servers is unreliable.**



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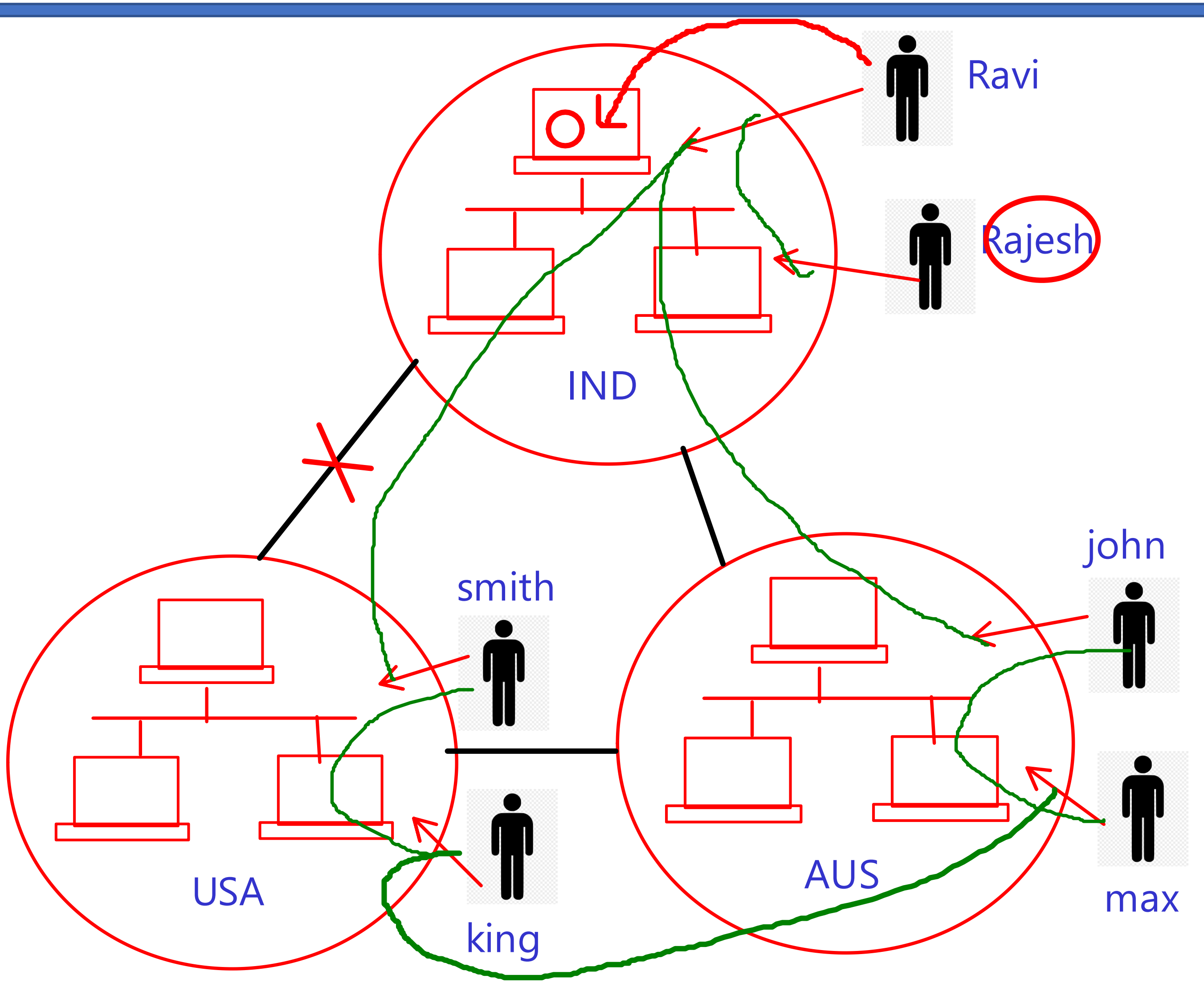

BASE Transaction

BA => Basically Available

system running 24x7

S => soft state

Data is auto transferred to all the node in cluster

E => Eventual consistancy

same data visible to all the clients eventually

Thank you!

Pradnyaa S. Dindorkar <pradnya@sunbeaminfo.com>